WORKED SOLUTION

STEP I 2003, Question 11

Mechanics • Collisions on an Inclined Plane

STEP I 2003, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q11

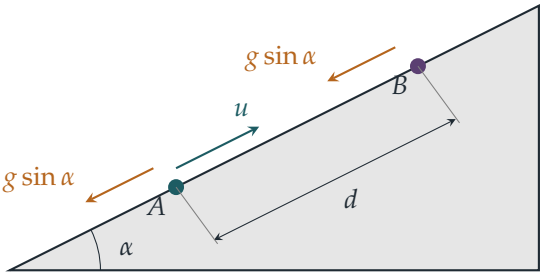
The key idea

On a smooth slope every particle has the same acceleration, $g\sin\alpha$ down the line of greatest slope, whatever its velocity. So gravity cannot change the *relative* motion of the two particles: seen from one of them, the other moves at constant velocity. That makes the collision time easy to find, and it tells us the relative velocity at impact without any messy kinematics. The energy loss in a collision depends only on that relative velocity, which is why α, g and d all disappear from the answer.

Solution

The model

We treat both particles as point masses moving along the line of greatest slope through A and B. The plane is smooth, so the only forces on each particle are its weight and the normal reaction. Resolving perpendicular to the plane, the reaction balances $mg\cos\alpha$. Resolving along the plane, each particle has acceleration $g\sin\alpha$ down the slope. Take x as the distance up the slope from A.



The time of collision

Let $a = g\sin\alpha$. With up the slope as positive, the particle from A (call it P) and the particle from B (call it Q) have positions

$$x_P = ut - \tfrac{1}{2}at^2,$$
$$x_Q = d - \tfrac{1}{2}at^2.$$

The gap between them is

$$x_Q - x_P = d - ut,$$

because the $\frac{1}{2}at^2$ terms cancel. The gap shrinks steadily from d and is zero exactly when

$$t = \frac{d}{u},$$

which is what we had to show. The cancellation is the key idea at work: both particles fall down the slope in the same way, so only the initial relative speed u matters.

The loss of kinetic energy

At $t = d/u$ the velocities up the slope are

$$v_P = u - \frac{ad}{u}, \qquad v_Q = -\frac{ad}{u}.$$

Write $w = ad/u$ for short. Then $v_P = u - w$ and $v_Q = -w$, and P approaches Q with relative speed $v_P - v_Q = u$.

> **Ask yourself** Why can we use conservation of momentum when gravity is acting?

The collision is instantaneous. Over the tiny time of impact, the finite forces of gravity and the normal reaction produce a negligible impulse compared with the large impulse between the particles. So momentum along the slope is conserved in the collision.

Let the velocities up the slope after the collision be v'_P and v'_Q. Since the masses are equal, conservation of momentum gives

$$v'_P + v'_Q = v_P + v_Q = u - 2w. \tag{1}$$

Newton's experimental law gives the speed of separation as e times the speed of approach:

$$v'_Q - v'_P = e(v_P - v_Q) = eu. \tag{2}$$

Rather than solving for v'_P and v'_Q separately, we use the identity

$$p^2 + q^2 = \tfrac{1}{2}[(p+q)^2 + (p-q)^2].$$

Applying it before and after the collision, the total kinetic energies are

$$\text{KE before} = \tfrac{1}{2}m(v_P^2 + v_Q^2) = \tfrac{1}{4}m[(u-2w)^2 + u^2],$$
$$\text{KE after} = \tfrac{1}{2}m(v_P'^2 + v_Q'^2) = \tfrac{1}{4}m[(u-2w)^2 + e^2u^2],$$

using (1) and (2) for the sum and difference after the collision. The sum term is the same in both, so

$$\text{loss of KE} = \tfrac{1}{4}mu^2 - \tfrac{1}{4}me^2u^2 = \tfrac{1}{4}mu^2(1-e^2),$$

as required.

Ask yourself What is the $(u-2w)^2$ term physically, and why does it not change?

Half of $v_P + v_Q$ is the velocity of the centre of mass. The term $\frac{1}{4}m(u-2w)^2$ is the kinetic energy of the total mass $2m$ moving with that velocity, and momentum conservation keeps it fixed. Only the kinetic energy of the motion *relative* to the centre of mass can be lost, and that depends only on the relative speed, u before and eu after.

Checks

- If $e = 1$ (perfectly elastic) the loss is zero, as it should be. If $e = 0$ the particles move together afterwards and the loss is $\frac{1}{4}mu^2$, the full kinetic energy of the relative motion.
- If $\alpha = 0$ the plane is horizontal, Q stays at rest, and P reaches it at speed u after time d/u, which agrees.
- Numerically, with $\alpha = 30^\circ$, $g = 9.8$, $d = 2$, $u = 4$, $e = \frac{1}{2}$ and $m = 1$: at $t = 0.5$ the velocities are 1.55 and -2.45. Afterwards they are -1.45 and 0.55, and the kinetic energy falls from 4.2025 to 1.2025, a loss of 3, which equals $\frac{1}{4} \cdot 16 \cdot \frac{3}{4}$. A random test of 1000 cases, stepping the motion numerically, agreed too.

Common slips

- Taking Q's velocity at impact to be zero. It has been sliding down the slope for time d/u, just as P has been slowing down.
- Getting a sign wrong in Newton's law. Write both velocities in the same positive direction, then separation speed $= e \times$ approach speed.
- Solving fully for v_P' and v_Q' and then expanding four squares. It works, but the sum-and-difference identity avoids almost all the algebra.

Beyond STEP

Both parts are examples of changing to the centre-of-mass frame. A uniform gravitational field gives every particle the same acceleration, so it drops out of relative motion completely. Also, the kinetic energy of any system splits into the kinetic energy of the centre of mass plus the kinetic energy relative to it (König's theorem). In two-body

problems the second part is $\frac{1}{2}\mu v_{\text{rel}}^2$ with the reduced mass $\mu = m_1 m_2/(m_1 + m_2)$, here $\frac{1}{2}m$. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both ideas carefully.

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